FinCompass: Validation-Gated Probabilistic Forecasting

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Abstract

A probability forecast in a financial market is a claim about an event that has not yet occurred, made with information that was available at a particular time. That simple statement creates several practical difficulties. Financial labels overlap, company information arrives asynchronously, macroeconomic series are revised, the same data are often examined repeatedly, and a model that was calibrated during development may become unreliable after deployment. This paper develops a forecasting architecture in which those difficulties are part of the model specification rather than after-the-fact warnings. Three quantities are kept separate. The first is a Bayesian evidence score used for interpretable research triage; its uncertainty concerns the score itself and is not a return probability. The second is a frozen empirical probability of a fixed forward excess-return event. It is trained with point-in-time inputs, purge-and-embargo rules, separated calibration and stacking stages, and a locked test set. The third is a bounded online Bayesian residual that reacts to new market, filing, and macroeconomic information while leaving the validated anchor unchanged. Online parameters are updated only after the original forward outcome has matured. The adaptive correction is applied only after date-balanced prequential loss, calibration, temporal breadth, freshness, and drift conditions are satisfied. The paper gives the mathematical specification, including fractional-Beta evidence aggregation, Bayesian logistic inference with a Laplace approximation, constrained probabilistic stacking, dependence-aware bootstrap validation, freshness decay, sequential rank-one covariance updating, and fail-closed deployment logic. In addition to synthetic regression fixtures, which remain permanently ineligible for market use, the implementation includes a real historical 12-month monthly-frequency ensemble that passed its configured locked-test research gate: ROC AUC 0.5762, Brier skill 0.0364, log-loss skill 0.0270, ECE 0.0447, and calibration slope 1.2723 on 532 locked-test samples across 76 dates. The artifact is classified as `validated_research`, not `validated_market`, because survivorship, delisting, and corporate-action controls are not established at the stronger level. This separation between statistical eligibility and data-governance strength is central to the framework.

Keywords: probabilistic forecasting; Bayesian logistic regression; online learning; calibration; prequential evaluation; temporal leakage; concept drift; point-in-time data; financial machine learning.

1 Introduction

Suppose a forecasting system assigns a stock a probability of 0.58 of outperforming a benchmark over the next year. At noon the company files a material report, the stock gaps relative to the benchmark, and credit spreads move at the same time. There are two unsatisfactory ways to react. A purely static model ignores the information until the next scheduled retraining. An uncontrolled online model can react immediately, but it can also begin "learning" from the new observation before the one-year outcome is known. The first approach is slow; the second can quietly destroy the meaning of the original validation.

The practical question is therefore not simply how to predict. It is how to change a probability when the information set changes, without pretending that the future label has already arrived. A second question follows immediately: how should a forecasting system demonstrate that its probabilities deserve to be interpreted as probabilities rather than as scores with a percent sign attached?

These questions are unusually difficult in financial data. The signal-to-noise ratio is low, many observations share the same market environment, long-horizon labels overlap, and researchers can test many alternatives on the same finite history. The probability of obtaining an apparently attractive backtest therefore increases with repeated experimentation, even when no stable economic relation has been found. This problem has been studied from several directions, including data snooping, multiple testing and backtest overfitting (White, 2000; Harvey and Liu, 2020; Bailey et al., 2017). Temporal leakage adds another layer: if an observation dated t uses a filing, revision or price that was not actually knowable at t, the backtest is no longer a historical simulation of the information available to the forecaster.

The approach developed here is conservative by construction. It separates three statistical objects that are often mixed together in financial applications. A research score summarizes current evidence and carries uncertainty about that score. A frozen anchor model estimates an empirical forward-event probability and is not changed casually after validation. A third model learns a bounded residual correction from newly arriving information. The residual may alter a candidate probability immediately, but its coefficients are updated only when the corresponding outcome has matured. If the online layer has too little temporal evidence, poor calibration, worse proper scores, stale inputs, or a drift alert, the system falls back to the frozen anchor.

The distinction matters. A statement such as “there is a high posterior probability that the research score exceeds 8” is not the same statement as “there is a 64% probability that the stock will outperform.” The former is uncertainty about an internally defined score. The latter is an empirical forecasting claim that must be calibrated against realized events. Keeping these claims apart is the starting point of the model.

The reference implementation used to make the discussion concrete is FinCompass. The software is not the contribution by itself. It is useful here because every equation in the paper is tied to an executable rule, a validation gate, or an audit artifact. Synthetic regression fixtures remain permanently ineligible for live market use. Separately, a real historical monthly model is retained as a `validated_research` artifact after passing its configured locked-test gate. Its role is deliberately bounded: it provides an operational pre-trained anchor for research forecasting and Live activation, while its data-governance limitations prevent promotion to `validated_market`.

1.1 How to read the paper

The paper is written in layers. Sections 2–4 can be read without a background in Bayesian statistics. They define the event being forecast and explain why the three probabilities must remain separate. Sections 5–10 give the full statistical model and validation protocol. Sections 11–13 deal with online activation, drift, lineage and reproducibility. The appendices collect derivations and implementation defaults for readers who want to reproduce the calculations exactly.

2 Financial probabilities

A useful probability forecast must answer three questions: what event is being forecast, what information was available when the forecast was made, and how often similar probabilities have been correct in held-out data. Without those three pieces, a number between zero and one is not yet a defensible probability forecast.

Let i denote a security, b a benchmark, and t an observation time. Let $\mathcal{F}_t$ denote all information available to the system at time t. A valid feature vector $X_{i,t}$ must be measurable with respect to $\mathcal{F}_t$; informally, every input must have been knowable at the time of the forecast.

For a horizon of H common trading sessions, define the stock and benchmark returns

$$R_{i,t}^{(H)} = \frac{P_i(\tau_H(t))}{P_i(t)} - 1, \tag{1}$$

$$R_{b,t}^{(H)} = \frac{P_b(\tau_H(t))}{P_b(t)} - 1, \tag{2}$$

where $\tau_H(t)$ is the Hth trading session on which both the stock and benchmark have a usable price after t. For an excess-return hurdle η, the binary event is

$$Y_{i,t}^{(H,\eta)} = \mathbb{I}\left\{R_{i,t}^{(H)} - R_{b,t}^{(H)} > \eta\right\}. \tag{3}$$

A daily annual target can be expressed with $H = 252$ common trading sessions and $\eta = 0$. The bundled real historical research model uses an equivalent 12-month event on monthly closes and compares the asset with the S&P 500 index. The target frequency is therefore part of model identity rather than an interchangeable display setting.

Equation (3) is more important than it may first appear. The outcome is not "whatever happened about a year later." It is attached to a precise endpoint. If the background process that resolves labels is delayed by two weeks, the target does not move by two weeks. Let

$$\mathcal{T}_{i,b}(t) = \{u_1 < u_2 < \cdots\} \tag{4}$$

be the ordered common trading sessions after t. Then

$$\tau_H(t) = u_H. \tag{5}$$

A label is not available until u_H exists, and once it exists the endpoint is u_H, not the date on which a scheduler happens to run.

The corresponding forecasting probability is

$$p_{i,t} = \mathbb{P}\left(Y_{i,t}^{(H,\eta)} = 1 \mid \mathcal{F}_t\right). \tag{6}$$

A model may estimate this probability badly, of course, but Equation (6) states what the number is intended to mean.

2.1 A small numerical example

Consider an anchor probability $p^A = 0.58$. Its log-odds are

$$\operatorname{logit}(0.58) = \log\frac{0.58}{0.42} \approx 0.323. \tag{7}$$

Suppose new information produces a validated residual correction of 0.33 in log-odds units. The candidate probability is then

$$p^C = \operatorname{sigmoid}(0.323 + 0.33) \approx 0.658. \tag{8}$$

The candidate has moved from 58% to about 65.8%. But this does not imply that the residual model is allowed to control the live output. If the online validation gate is still warming up, the live probability remains 0.58. The event can change the candidate belief now; the system learns from whether that belief was useful only after the future outcome becomes observable.

This simple example contains most of the architecture. The rest of the paper makes each step precise.

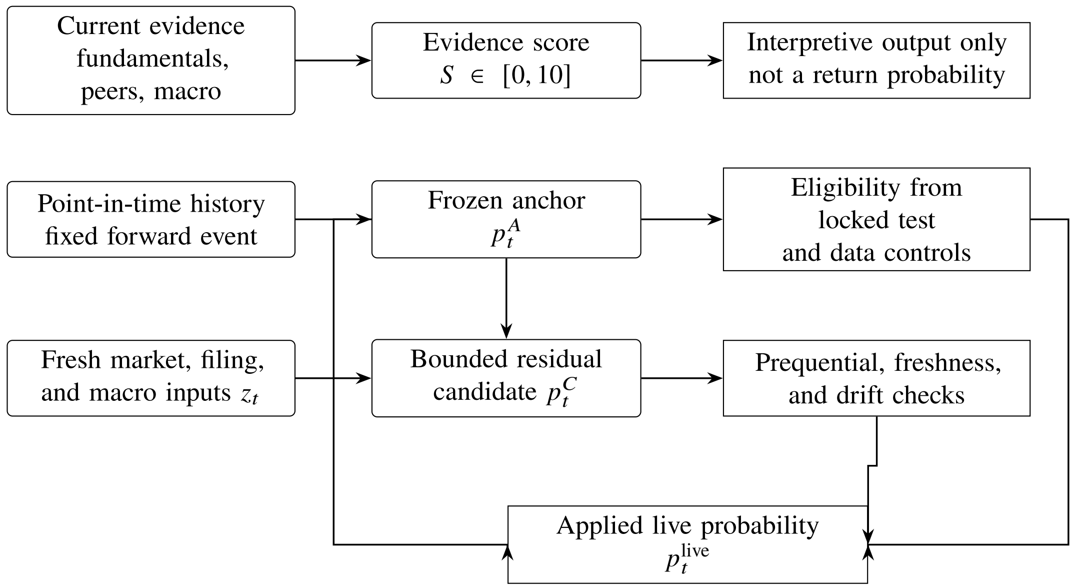

Figure 1: The three statistical objects used by FinCompass have different meanings and different operational rules. A research score is interpretive, the anchor probability is the validated forecast object, and the live probability is a bounded update around that anchor.

3 Three statistical objects

The architecture is easiest to understand if the three outputs are named before the mathematics is introduced.

The *evidence score* is a 0–10 research summary built from profitability, durability, safety, valuation and current-regime evidence. Its posterior distribution expresses uncertainty about that score. It is not trained to match frequencies of future outperformance.

The *anchor probability* is an empirical model of Equation (3). It is trained on historical observations that respect point-in-time availability and is frozen after validation. It can be live only if its data and locked-test results satisfy a defined governance tier.

The *adaptive probability* starts from the anchor and applies a bounded residual in log-odds space using fresh market, filing and macroeconomic information. The residual has its own posterior state, performance history and activation gate.

Figure 1 shows the separation. The arrows indicate permissible information flow; they do not imply that every plane is eligible for live use.

A common failure in financial software is to collapse these layers. A high research score is relabeled as a high probability of return, or a live price shock is fed into a model that was validated only on end-of-day data. The present architecture prohibits both operations unless the corresponding probability model has been validated for the claimed information contract.

4 Point-in-time information

The phrase *point in time* means more than attaching a date to a row. A historical observation should contain only values that a forecaster could have possessed on that date.

For an accounting item whose fiscal period ends at q, let $a(q)$ be its public availability time. The item is eligible for a feature vector observed at time t only when

$$a(q) \leq t. \tag{9}$$

Joining fundamentals by fiscal period end alone can therefore create hindsight. A March quarter that was filed in May must not appear in an April observation.

The same issue occurs in macroeconomic data. Many series are revised. A backtest that fills every historical date with the latest revised value uses information that did not exist in real time. Vintage-aware databases such as ALFRED are designed precisely to distinguish the value known today from the value known at an earlier real-time date (Federal Reserve Bank of St. Louis, 2026). For company filings, the SEC disseminates submission information throughout the day; in the live layer, the acceptance timestamp is used when available, with the filing date only as a fallback (SEC, 2026).

Prices require their own discipline. The anchor model retains the same daily feature semantics used during its validation. An intraday quote is not substituted into a daily feature merely because the quote is newer. Intraday information belongs to the separately validated residual layer.

The point-in-time contract can be summarized by one rule: if the historical state of the system could not have known a value at t, the value does not belong in $X_{i,t}$.

5 The evidence plane

The first plane is deliberately simpler than the forecasting model. It answers a research question: given the evidence presently available, how strong is the company-and-regime profile under a transparent scoring rubric, and how uncertain is that score because evidence is incomplete?

This plane uses five pillars: quality, financial durability, safety, valuation and current cycle/regime context. Each raw metric is first mapped to a continuous 0–10 scale. The continuity matters because a company with a return on invested capital of 11.99% should not receive a materially different score from an otherwise identical company at 12.01% merely because a hand-built bin boundary was crossed.

5.1 Continuous metric maps

Suppose metric x is represented by ordered scoring knots (a_k, s_k), where a_k is a raw metric value and $s_k \in [0, 10]$. Between adjacent knots, the score is linearly interpolated:

$$f(x) = s_k + \frac{x - a_k}{a_{k+1} - a_k}(s_{k+1} - s_k), \qquad a_k \leq x \leq a_{k+1}. \tag{10}$$

Values beyond the outer knots are clipped to the endpoint scores. The function f is chosen to be economically monotone over the region where such monotonicity is defensible.

When a sufficiently large sector peer sample is available, the absolute score can be supplemented with a robust relative measure. Let m_j and IQR_j be the peer median and interquartile range for metric j. Define

$$z_j^{\mathrm{IQR}} = d_j \frac{x_j - m_j}{\max(\mathrm{IQR}_j, \varepsilon)}, \tag{11}$$

where $d_j = 1$ when larger values are preferable and $d_j = -1$ when smaller values are preferable. The peer score is

$$s_j^{\mathrm{peer}} = \mathrm{clip}\left(5 + 2.2 z_j^{\mathrm{IQR}}, 1, 9.5\right). \tag{12}$$

With peer weight ρ, the metric score is

$$s_j = (1 - \rho) s_j^{\mathrm{abs}} + \rho s_j^{\mathrm{peer}}. \tag{13}$$

The default implementation uses $\rho = 0.25$ for the general case, with small variations for specific metrics. If fewer than five suitable peers are available, the relative term is omitted instead of being estimated from an unstable sample.

5.2 Why a Beta model appears here

The 0–10 score can be normalized as $q_j = s_j/10$. This is not a Bernoulli observation in a literal frequentist sense. It is fractional evidence on a bounded scale. A Beta distribution is useful because it provides a transparent way to accumulate such evidence while shrinking sparse pillars toward a neutral prior.

For pillar r, let $\mathcal{J}_r$ be the available metrics and let w_j be their evidence weights. Starting from prior $\mathrm{Beta}(\alpha_0, \beta_0)$, the fractional update is

$$\alpha_r = \alpha_0 + c \sum_{j \in \mathcal{J}_r} w_j q_j, \tag{14}$$

$$\beta_r = \beta_0 + c \sum_{j \in \mathcal{J}_r} w_j (1 - q_j), \tag{15}$$

where c controls the strength assigned to one unit of rubric evidence. The default prior is neutral, $\alpha_0 = \beta_0 = 2$, and the implementation uses $c = 4$.

If $U_r \sim \mathrm{Beta}(\alpha_r, \beta_r)$, the reported pillar score is

$$S_r = 10\mathbb{E}[U_r] = 10\frac{\alpha_r}{\alpha_r + \beta_r}. \tag{16}$$

The posterior variance is

$$\mathrm{Var}(U_r) = \frac{\alpha_r \beta_r}{(\alpha_r + \beta_r)^2(\alpha_r + \beta_r + 1)}. \tag{17}$$

When little evidence is present, the posterior remains close to the neutral prior and its interval remains broad. With more consistent evidence, the posterior moves away from five and becomes narrower. This is the desired behavior: missing data reduce confidence rather than being silently replaced by favorable assumptions.

Evidence coverage is reported separately. If W_r^{obs} is the sum of observed metric weights and W_r^{exp} is the expected weight under the pillar definition, then

$$C_r = \min\left(1, \frac{W_r^{\mathrm{obs}}}{W_r^{\mathrm{exp}}}\right). \tag{18}$$

A score of 8 with low coverage is therefore distinguishable from a score of 8 supported by a nearly complete evidence set.

5.3 Composite uncertainty and overlapping pillars

Let λ_r be the fixed pillar weights, with $\sum_r \lambda_r = 1$. A naive composite draw would be

$$S^{(m)} = 10 \sum_r \lambda_r U_r^{(m)}. \tag{19}$$

The difficulty is that the pillars are not independent. Quality and durability can share profitability evidence, and cycle context can reuse valuation or capital-return information. Treating all pillar posteriors as independent would make the composite interval artificially narrow.

Rather than inventing an empirical covariance matrix that has not been estimated, the implementation propagates two dependence scenarios. In the first, pillar draws are independent. In the second, the marginal draws are sorted and paired by rank, creating a comonotonic, perfectly positively dependent scenario. If $[L_{\text{ind}}, U_{\text{ind}}]$ and $[L_{\text{com}}, U_{\text{com}}]$ are the corresponding 90% intervals, the reported model interval is the envelope

$$[L, U] = [\min(L_{\text{ind}}, L_{\text{com}}),\ \max(U_{\text{ind}}, U_{\text{com}})] . \tag{20}$$

This is a sensitivity analysis, not a claim that the real dependence is comonotonic. Its purpose is to prevent a convenient independence assumption from communicating more precision than the evidence supports.

The current-regime pillar also carries a causal guardrail. It may use valuation regime, capital-return quality, yield-curve slope, high-yield spreads and commodity context, but it is described only as current regime evidence. It is not interpreted as a recession timer or a forward return forecast. That distinction is enforced in the user-facing explanation and by regression tests in the reference implementation.

6 The frozen anchor

The anchor is the first plane that is allowed to make the probability claim in Equation (6). It is not derived from the 0–10 evidence posterior. It is fitted to labeled historical events.

The reference ensemble contains three deliberately different learners: Bayesian logistic regression, histogram gradient boosting, and a regularized random forest. The logistic component provides interpretable posterior coefficient uncertainty. The two tree-based models allow nonlinearities and interactions that a single linear log-odds surface cannot capture. The models are calibrated separately, combined with nonnegative weights, and calibrated once more at the ensemble level.

6.1 Bayesian logistic component

Let $x_n \in \mathbb{R}^p$ be a feature vector and $y_n \in \{0, 1\}$ its target. With an intercept included in x_n, the logistic model is

$$\mathbb{P}(Y_n = 1 \mid x_n, \beta) = \operatorname{sigmoid}(x_n^\top \beta), \qquad \operatorname{sigmoid}(a) = \frac{1}{1 + e^{-a}}. \tag{21}$$

A zero-centered Gaussian prior is placed on the coefficients,

$$\beta \sim \mathcal{N}(0, \sigma_\beta^2 I). \tag{22}$$

The strict default uses $\sigma_\beta = 1.5$ after the feature preprocessing defined by the model pipeline.

Ignoring constants, the log posterior is

$$\ell(\beta) = \sum_{n=1}^{N} [y_n \log p_n + (1 - y_n) \log(1 - p_n)] - \frac{1}{2\sigma_\beta^2} \beta^\top \beta, \tag{23}$$

where $p_n = \operatorname{sigmoid}(x_n^\top \beta)$. The maximum a posteriori estimate $\hat{\beta}$ maximizes Equation (23). The negative Hessian at the mode is

$$H = X^\top W X + \sigma_\beta^{-2} I, \qquad W = \operatorname{diag}\{p_n(1 - p_n)\}. \tag{24}$$

A Laplace approximation gives

$$\beta \mid \mathcal{D} \approx \mathcal{N}(\hat{\beta}, H^{-1}). \tag{25}$$

Posterior coefficient draws can then be propagated through Equation (21) to obtain a coefficient-uncertainty interval for the logistic component (MacKay, 1992).

The interval is useful but should not be oversold. It conditions on the chosen model family and training data; it does not include every form of model misspecification, structural break, or market uncertainty.

6.2 Nonlinear components and component calibration

The histogram gradient-boosting and random-forest components are standard probabilistic classifiers (Friedman, 2001; Breiman, 2001). Their raw class probabilities are not assumed to be calibrated. Each component is therefore mapped through a calibration function using a chronologically later validation stage.

For sigmoid calibration, a convenient form is

$$C(p) = \mathrm{sigmoid}\{a \ \mathrm{logit}(p) + b\}. \tag{26}$$

The implementation also supports isotonic calibration, a monotone nonparametric mapping, but the strict reference profile uses sigmoid calibration. Calibrators are fit without using the locked test set (Niculescu-Mizil and Caruana, 2005; Gneiting et al., 2007).

6.3 Constrained probabilistic stacking

Let $\tilde{p}_{nk}$ be the calibrated probability from component k for validation observation n. The uncalibrated ensemble probability is

$$q_n = \sum_{k=1}^{K} w_k \tilde{p}_{nk}, \qquad w_k \geq 0, \quad \sum_{k=1}^{K} w_k = 1. \tag{27}$$

The weights are selected on a separate chronological validation stage by minimizing a regularized Brier objective,

$$\hat{w} = \arg \min_{w \in \Delta_K} \left\{ \frac{1}{N} \sum_{n=1}^{N} (q_n - y_n)^2 + \lambda \sum_{k=1}^{K} \left(w_k - \frac{1}{K} \right)^2 \right\}, \tag{28}$$

where Δ_K is the probability simplex and the implementation uses $\lambda = 0.01$. A final calibrator C_E is fitted on a third chronological validation stage:

$$p_n^A = C_E(q_n). \tag{29}$$

This separation is intentional. If component calibration, stacking and final calibration are optimized on the same observations, the reported calibration can be better than the procedure deserves.

7 Temporal validation

A conventional random train/test split assumes that observations can be exchanged without changing the problem. Long-horizon financial labels violate that assumption. If forecasts are sampled monthly but evaluated one year ahead, neighboring observations share much of the same future return path. Securities observed on the same date also share market conditions.

The validation design therefore has two jobs: prevent information from crossing chronological boundaries and preserve enough dependence in uncertainty estimates that the effective sample size is not exaggerated.

7.1 Purge and embargo

Suppose a later partition begins at date s. A training observation at date t is retained only if its target has fully resolved before s,

$$\tau_H(t) < s. \tag{30}$$

An additional embargo of E business days requires

$$t < s - \mathrm{BDay}(E). \tag{31}$$

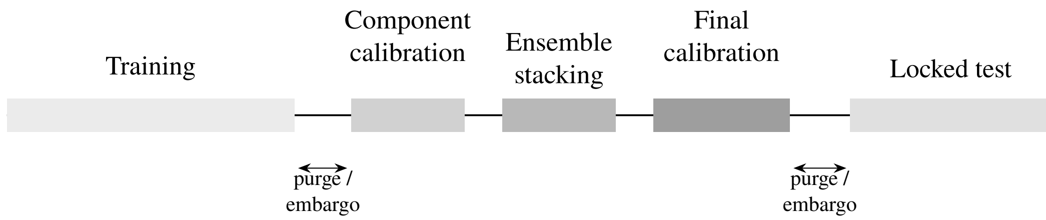

Figure 2: Chronological development protocol. Training, calibration, stacking, and final locked testing are separated in time so that the locked test remains a true confirmatory check.

The strict reference configuration uses $E = 252$ at the broad train-to-validation and validation-to-test boundaries. The three internal validation roles used for component calibration, stacking and final calibration remain chronological and purge any observation whose forward target crosses the next internal stage boundary, but they do not reapply the full outer embargo. Reapplying a one-year embargo inside a much smaller validation partition can eliminate an entire fitting stage without adding a new leakage barrier.

Figure 2 shows the order. The large blank gaps at the outer partition boundaries represent purge plus embargo. The internal validation roles are also separated chronologically, but use forward-target purging rather than a repeated full outer embargo.

The locked test has a strict meaning: it is not used to choose features, hyperparameters, ensemble weights, calibration method, validation thresholds, or governance rules. If it is inspected repeatedly while those decisions are still changing, it is no longer a locked test. This is the same broad problem addressed by the backtest-overfitting literature: repeated search turns apparently out-of-sample evidence into another form of in-sample optimization (Bailey et al., 2017).

7.2 Moving date-block and cross-sectional cluster bootstrap

Even a clean locked test can understate uncertainty if its dependence structure is ignored. Let $d_1, \ldots, d_D$ denote the distinct observation dates in the test set, and let $C(d)$ contain all securities observed on date d. The bootstrap samples contiguous blocks of observation dates and includes the entire cross-sectional cluster $C(d)$ whenever date d is selected.

If forecasts are sampled every Δ trading days and evaluated over H trading days, the automatic block length is

$$L = \left\lceil \frac{H}{\Delta} \right\rceil. \tag{32}$$

For $H = 252$ and $\Delta = 21$, $L = 12$ observation dates. This combines the logic of block resampling for serial dependence with same-date clustering for market-wide dependence (Künsch, 1989; Politis and Romano, 1994). It does not prove that the bootstrap captures every dependence mechanism; it is a deliberately more conservative alternative to treating each row as an independent draw.

8 Scoring probabilistic forecasts

Classification accuracy throws away most of the information in a probability. A forecast of 0.51 and a forecast of 0.99 produce the same class label if both are thresholded at 0.5, even though the second makes a much stronger claim. Proper scoring rules retain that distinction.

For observations (p_n, y_n), the Brier score is

$$\mathrm{BS} = \frac{1}{N}\sum_{n=1}^{N}(p_n - y_n)^2, \tag{33}$$

and binary log loss is

$$\mathrm{LL} = -\frac{1}{N}\sum_{n=1}^{N}\left[y_n \log p_n + (1 - y_n)\log(1 - p_n)\right]. \tag{34}$$

Both reward honest probability estimates and penalize confident errors (Brier, 1950; Gneiting and Raftery, 2007).

A reference model uses an event rate estimated from development data, not from the locked test. For either score,

$$\mathrm{Skill} = 1 - \frac{\mathrm{Score}_{\mathrm{model}}}{\mathrm{Score}_{\mathrm{reference}}}. \tag{35}$$

Positive skill means the model improves on the constant-probability reference under that proper score.

Discrimination and calibration are checked separately. ROC AUC and average precision describe ranking behavior. Calibration asks whether events with forecast probability near p occur at approximately frequency p. A calibration regression may be written

$$\operatorname{logit}\mathbb{P}(Y = 1) = a + b\operatorname{logit}(p). \tag{36}$$

Ideal calibration corresponds to $a = 0$ and $b = 1$. Expected calibration error (ECE) gives a binned summary, but it must be interpreted with care because its value depends on binning and sample composition.

The strict fixture gate requires, among other conditions, positive Brier and log-loss skill, a minimum AUC, acceptable ECE and calibration slope, adequate class counts, enough locked-test dates and elapsed span, positive skill in a sufficient fraction of purged walk-forward folds, and dependence-aware bootstrap bounds that do not cross specified failure thresholds. A passing gate is evidence about a particular model, target, dataset and protocol. It is not a general certificate that the market relation will persist indefinitely.

9 Anchor uncertainty

The Bayesian logistic component provides a natural distribution over coefficients. The other components do not. The implementation therefore reports an *uncertainty envelope*, not a formal predictive interval with guaranteed market coverage.

For each observation, Bayesian coefficient draws produce lower and upper logistic-component probabilities. These are calibrated. The ensemble also has calibrated component point probabilities from gradient boosting and random forest. The lower envelope is the minimum of the Bayesian lower bound and all component probabilities; the upper envelope is defined analogously. Because the final sigmoid or isotonic calibrator is monotone, the envelope endpoints can be passed through the final calibrator.

The result is useful for abstention and model-disagreement diagnostics, but its interpretation is deliberately narrow: it combines Bayesian coefficient uncertainty and inter-model dispersion. It does not integrate every uncertainty source in the data-generating process. In particular, it is not a prediction interval for a future return.

The reference implementation may abstain when the final probability lies sufficiently close to 0.5 or when an interval rule configured by the user is triggered. Abstention is not a failure to predict; it is an explicit acknowledgement that a binary decision would be more precise than the model evidence warrants.

10 The adaptive residual

A validated anchor should not be refitted every time new information appears. At the same time, a live system should not ignore a material filing or a large relative price move simply because the next full retraining is weeks away. The compromise used here is a residual model in log-odds space.

Let p_t^A be the frozen anchor probability and let $z_t \in \mathbb{R}^{13}$ be the live feature vector. The adaptive coefficient state at time t has posterior mean m_t and covariance P_t. The raw residual is

$$\delta_t^{\text{raw}} = z_t^\top m_t. \tag{37}$$

To prevent the residual from overwhelming the anchor, the log-odds shift is bounded:

$$\delta_t = \text{clip}\left(\delta_t^{\text{raw}}, -d_{\max}, d_{\max}\right). \tag{38}$$

The balanced reference profile uses $d_{\max} = 0.75$. The candidate probability is

$$p_t^C = \text{clip}\left[\text{sigmoid}\left\{\text{logit}(p_t^A) + \delta_t\right\}, \epsilon, 1-\epsilon\right], \tag{39}$$

with $\epsilon = 0.01$ in the balanced profile.

There is a useful interpretation of the bound. Since odds are multiplied by e^{δ_t}, the residual can change the anchor odds by at most

$$\exp(d_{\max}) = \exp(0.75) \approx 2.12 \tag{40}$$

in either direction before probability clipping. The online layer is therefore influential, but it is not allowed to erase the validated anchor in one step.

The posterior variance of the unbounded log-odds correction is

$$s_{\delta,t}^2 = z_t^\top P_t z_t. \tag{41}$$

This quantity is displayed as a state-uncertainty diagnostic. As with the anchor envelope, it should not be confused with a complete predictive distribution for market returns.

10.1 The 13-dimensional live feature contract

The implemented vector is

$$\begin{aligned} z_t = (&r_{1d},\ r_{1d}^{\text{rel}},\ z_{r,20},\ z_{r,20}^{\text{rel}},\ z_{V,20},\ \text{range}_{1d}, \\ &f_{\text{SEC}},\ f_{8K/6K},\ f_{10Q},\ f_{10K},\ \Delta YC,\ \Delta HY,\ f_{\text{macro}})^\top. \end{aligned} \tag{42}$$

The first six features describe current market behavior. Four features represent the freshness and broad family of the most recent SEC filing. Two represent changes in the yield curve and high-yield spread. The final term measures macroeconomic freshness.

Table 1 gives the contract in words. The feature names are intentionally modest. For example, a fresh 10-Q indicator says that a quarterly filing arrived recently; it does not claim that the filing is good or bad. A future extension may add validated textual or numerical event interpretation, but an external feed is context-only by default until separately validated.

Table 1: Adaptive feature vector used by the reference implementation. Inputs are grouped so that each term has a clear operational meaning in the live residual.

Feature	Interpretation
Market return, 1 day	Previous session close to latest price; includes the overnight gap when available.
Benchmark-relative return, 1 day	Stock move minus benchmark move over the corresponding interval.
Return z-score, 20	Standardized recent stock return shock.
Relative-return z-score, 20	Standardized recent stock-minus-benchmark shock.
Volume z-score, 20	Standardized unusual volume.
Intraday range	High-low range normalized by price.
SEC filing freshness	Exponentially decayed age of the most recent filing.
8-K/6-K, 10-Q, 10-K-family terms	Filing-family indicator multiplied by SEC freshness.
Yield-curve change	Recent curve change, freshness-weighted.
High-yield spread change	Recent spread change, freshness-weighted.
Macro freshness	Exponentially decayed age of the current macro observation.

10.2 Freshness is different from event age

A live system must distinguish an old event from an unverified source. If no new 10-Q has been filed for several weeks, the most recent filing is simply old. If the SEC source has not been successfully checked for several days, the system cannot know whether a newer filing was missed.

For a source event of age a seconds and half-life h hours, the freshness multiplier is

$$f(a; h) = \exp\left(-\log(2)\frac{a}{3600h}\right). \tag{43}$$

The balanced profile uses a 36-hour event half-life for SEC information and four times that half-life for macroeconomic event decay. Separately, each provider has a maximum allowed verification staleness. If the most recent successful provider check is older than that limit, source-dependent adaptive features are set to zero until verification resumes. Thus an old event decays; an unverified source is suppressed.

The polling scheduler uses the timestamp of the last provider check, not the timestamp of the last distinct event. Otherwise a provider that repeatedly reports "nothing new" could be polled on every browser refresh after the first cadence window.

11 Online Bayesian learning

The adaptive model follows a predict-before-update rule. At observation time t, the system records p_t^A, z_t and the candidate p_t^C. It does not yet know Y_t. The pending label is bound to the base model, benchmark, target definition, adaptive settings fingerprint and state lineage. Only when the exact endpoint in Equation (5) is available is the outcome resolved and used for learning.

This ordering is a form of prequential evaluation: predict first, observe the outcome later, then update (Dawid, 1984). It prevents the model from evaluating a forecast with information obtained after the forecast was made.

11.1 Dynamic prior between updates

Let the posterior before a new matured observation be approximately

$$\theta_{t-1} \mid \mathcal{D}_{t-1} \approx \mathcal{N}(m_{t-1}, P_{t-1}). \tag{44}$$

Before incorporating the next label, uncertainty is inflated by a forgetting factor $\lambda_f \in (0, 1]$ and process noise $q \geq 0$:

$$P_t^- = \frac{P_{t-1}}{\lambda_f} + qI, \qquad m_t^- = m_{t-1}. \tag{45}$$

The balanced profile uses $\lambda_f = 0.997$ and $q = 0.0005$. Old information is therefore discounted gradually rather than discarded abruptly. This is related to dynamic logistic models in which coefficients are allowed to evolve over time (Penny and Roberts, 1999; McCormick et al., 2012).

11.2 Rank-one posterior update

For the matured observation, let $z = z_t$ and let $p = p_t^C$ be the probability issued before learning from its label y. The local Bernoulli curvature is

$$w = p(1 - p). \tag{46}$$

Define

$$v = P_t^- z, \qquad d = 1 + wz^\top v. \tag{47}$$

A one-step Gaussian/Laplace update is

$$P_t = P_t^- - \frac{w}{d} vv^\top, \tag{48}$$

$$m_t = m_t^- + \frac{v}{d}(y - p). \tag{49}$$

Equation (48) is the Woodbury/Sherman-Morrison form of the inverse Hessian update. It avoids recomputing a matrix pseudoinverse for every matured label. With only 13 adaptive coefficients, either method is feasible in isolation, but the rank-one form is numerically cleaner and has bounded cost for continuous use. A small positive-definiteness floor is applied after symmetrization if numerical roundoff produces an eigenvalue too close to zero.

The initial residual mean is zero. Consequently, before any online learning, the candidate probability equals the anchor probability. The prior covariance is $\sigma_0^2 I$, where the balanced profile uses $\sigma_0 = 0.75$.

11.3 Why repeated screen refreshes do not create evidence

A live application may display many predictions during one trading day. Those displays are not independent learning trials. The implementation permits frequent prediction refreshes but limits effective learning observations by ticker, model and observation date. More generally, the activation gate requires not only a minimum number of matured rows but also a minimum number of unique observation dates and a minimum elapsed calendar span.

This distinction becomes important in a large watchlist. Two hundred stocks observed on one date provide a large cross section but only one market date. They should not be mistaken for two hundred independent demonstrations that an online model is stable through time.

12 Prequential activation

The online model is allowed to produce a candidate from the first observation, but it is not automatically allowed to replace the anchor in the live output. Activation depends on recent matured forecasts.

Let $\mathcal{D}$ be the set of observation dates in the evaluation window and let n_d be the number of matured observations on date d. For a loss function ℓ, the date-balanced loss is

$$\bar{L} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \frac{1}{n_d} \sum_{i=1}^{n_d} \ell(p_{di}, y_{di}). \tag{50}$$

Every date receives the same total weight. This prevents a day with a large cross section from dominating the temporal evidence.

The same principle is used for online ECE. Give observation (d, i) weight

$$\omega_{di} = \frac{1}{|\mathcal{D}| n_d}. \tag{51}$$

For probability bin B_k, let $W_k = \sum_{(d,i)\in B_k} \omega_{di}$. Then

$$\mathrm{ECE}_{\mathrm{date}} = \sum_k W_k \left| \frac{\sum_{(d,i)\in B_k} \omega_{di} p_{di}}{W_k} - \frac{\sum_{(d,i)\in B_k} \omega_{di} y_{di}}{W_k} \right|. \tag{52}$$

This is not a new universal definition of ECE. It is a governance metric chosen to keep online calibration from being dominated by the number of tickers sampled on a particular date.

For the balanced profile, the gate requires at least 120 matured observations, at least 60 unique observation dates, and at least 180 calendar days of span. It also requires the adaptive date-balanced Brier score and log loss to be no worse than the anchor under zero-regret thresholds, ECE no greater than 0.08, and no drift alert. Formally, with checks $G_1, \ldots, G_m$,

$$G_t = \bigwedge_{j=1}^{m} G_j. \tag{53}$$

Only if $G_t = 1$ and adaptive application is enabled may the residual control the live probability.

13 Drift protection

A model can pass an activation gate and later deteriorate. The online layer therefore compares its squared-error loss with the anchor on each matured observation. Define

$$e_t = (p_t^C - y_t)^2 - (p_t^A - y_t)^2. \tag{54}$$

Positive e_t means the adaptive model was worse on that observation. An exponentially weighted mean and variance are maintained:

$$\mu_t = (1 - \alpha)\mu_{t-1} + \alpha e_t, \tag{55}$$

$$s_t^2 = (1 - \alpha)s_{t-1}^2 + \alpha(e_t - \mu_t)^2. \tag{56}$$

After a minimum number of samples, an alert is raised when

$$\mu_t > a + c\frac{s_t}{\sqrt{n_t}}, \tag{57}$$

where a is an allowance and c is a control multiplier. The balanced profile uses $\alpha = 0.08$, $a = 0.001$, $c = 2.75$, and a minimum of 60 samples.

This rule is a deterioration safeguard, not a formal hypothesis test with a claimed Type-I error rate. It borrows the practical logic of sequential control methods (Page, 1954; Roberts, 1959): persistent loss relative to the anchor is more important than one bad outcome. If the alert is active, the residual fails the gate and control returns to the anchor.

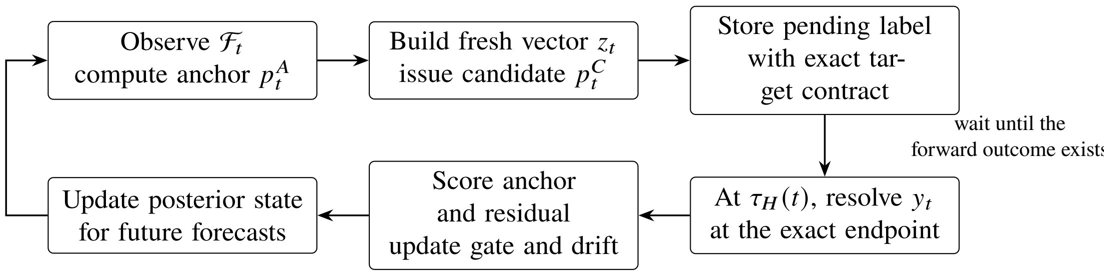

Figure 3: Predict-before-update cycle. New information may change a candidate probability immediately, but learning happens only after the original forward outcome is fully observed.

14 Fail-closed Live probability

The deployment rule can now be written compactly. Let A_t indicate that a frozen anchor is eligible for live use under its governance tier. Let G_t be the adaptive gate in Equation (53). Then

$$p_t^{\text{live}} = \begin{cases} \text{unavailable}, & A_t = 0, \\ p_t^C, & A_t = 1 \text{ and } G_t = 1, \\ p_t^A, & A_t = 1 \text{ and } G_t = 0. \end{cases} \tag{58}$$

The adaptive layer therefore cannot promote an ineligible anchor. It can only modify an anchor that has already earned live eligibility. Likewise, a source outage does not cause the system to infer that "nothing happened." Source-dependent residual features are suppressed when verification is stale.

Figure 3 shows the operational sequence for one forecast. Notice the long delay between prediction and learning when $H = 252$.

15 Lineage and reproducibility

A configurable research platform creates a subtle statistical problem: changing a setting can change the meaning of a model. A 126-day target is not the same model as a 252-day target. A QQQ-relative event is not the same event as an SPY-relative event. A residual trained with a responsive forgetting factor is not the same posterior state as one trained under a conservative profile.

For this reason, settings that change the target or learning semantics are part of model identity. A frozen model manifest records the target, feature contract, data provenance, split dates, calibration method, validation metrics and artifact hash. The model identifier is bound to the serialized artifact hash.

The adaptive state key is derived from the base model identifier, adaptive engine identifier and a fingerprint of the complete adaptive settings contract. Pending labels carry the same lineage. If a user changes from one adaptive profile to another, the new profile begins or resumes its own compatible state; it does not reinterpret the old posterior under new process noise, forgetting, freshness or gate assumptions.

This design has a practical benefit. A future audit can answer not only "what probability was shown?" but also "which target, model, settings, event state and validation tier produced it?" Reproducibility is therefore treated as part of the statistical model rather than as an optional software feature.

16 Validation tiers

Passing numerical thresholds is not sufficient to claim market validation if the dataset itself is biased. The reference implementation uses four tiers.

Table 2: Model governance tiers and what each tier permits the software to claim.

Tier	Meaning
`fixture_only`	Synthetic or deterministic data used to verify software, statistics, leakage controls and release regression behavior. Never live-eligible.
`rejected`	Required statistical or data-governance conditions were not met.
`validated_research`	The empirical model passed its locked-test gate on real historical data, but the dataset may still carry limitations that prevent the strongest market claim.
`validated_market`	In addition to statistical gates, the dataset documents point-in-time features, historical-universe/survivorship control, delisted securities, and corporate-action-adjusted prices.

Table 3: Frozen-anchor locked-test results on the synthetic fixture. These fixtures verify software behavior, not market validity.

Metric	Value
Brier score	0.217727
Brier skill	0.117815
Log loss	0.623959
Log-loss skill	0.091430
ROC AUC	0.699801
Average precision	0.623286
ECE (10 bins)	0.035934
Calibration slope	0.800691
Calibration intercept	-0.081872
Locked-test observations	2,304
Distinct test dates	72

The distinction between `validated_research` and `validated_market` prevents a common shortcut: reporting excellent test metrics on a present-day universe and overlooking the securities that disappeared from that universe. Delisting bias can materially distort historical performance (Shumway, 1997). A passing artifact is still only a candidate: operational activation is an explicit, separate action. The reference implementation never treats the newest usable artifact as live merely because it was created later.

17 Synthetic fixture results

The bundled fixtures answer a narrow but useful question: if the statistical code is changed, do the expected temporal splits, calibration stages, bootstrap procedures, online updates and gates still behave as intended? They do not answer whether the model contains exploitable market signal.

17.1 Frozen anchor fixture

The deterministic anchor fixture contains 6,528 training observations, 1,920 validation observations and 2,304 locked-test observations. The test partition spans 72 observation dates over 1,988 calendar days. Table 3 reports the locked-test metrics from the frozen reference artifact.

With the moving 12-date block/cross-sectional cluster bootstrap, the 90% lower bound for Brier skill is 0.10898, the lower bound for log-loss skill is 0.08343, the lower bound for ROC AUC is 0.69373, and the

Table 4: Adaptive locked-stream results on the synthetic fixture. The fixture remains permanently ineligible for live market use.

Metric	Frozen anchor	Adaptive	Improvement
Brier score	0.235658	0.204069	0.031588
Log loss	0.664106	0.596988	0.067118

upper bound for ECE is 0.04917. All four purged walk-forward folds have positive Brier skill; their median Brier skill is approximately 0.1303. The learned ensemble weights are approximately 0.4730 for Bayesian logistic regression, 0.2301 for histogram gradient boosting, and 0.2969 for random forest.

These values are deliberately easier to interpret than to market: they show that the fixture and validation machinery are internally consistent. Because the data are synthetic, none of the values is evidence that the same performance exists in real markets.

17.2 Adaptive streaming fixture

The adaptive fixture contains a 1,200-observation warm-up sequence followed by a 600-observation locked prequential stream. The model predicts before every update. The locked results are shown in Table 4.

At the end of the locked stream, the recent evaluation window contains 250 independent observation dates spanning 498 days. Its adaptive Brier score is 0.202803 versus 0.240625 for the anchor; its adaptive log loss is 0.595261 versus 0.674280; date-balanced ECE is 0.061924; and no drift alert is active. All configured balanced-profile checks pass.

Again, the artifact remains `fixture_only`. A software test is successful precisely because the software refuses to turn these synthetic statistics into a market claim.

18 Real historical 12-month model

Synthetic fixtures are necessary for deterministic regression testing, but they are not sufficient for evaluating a real historical probability model. A separate monthly-frequency ensemble was therefore trained on a real historical sample covering eight equities and the S&P 500 index. The target is whether the asset's monthly-close return over the next 12 months exceeds the S&P 500 monthly-close return over the same interval.

The model uses 20 monthly features covering own-asset return, relative return, benchmark return, volatility, drawdown, distance from a 12-month high, and a 3-to-12-month moving-average ratio. The component learners are Bayesian logistic regression, histogram gradient boosting, and random forest. Their fitted stacking weights are approximately 0.2799, 0.2295, and 0.4906, respectively. Component probabilities are calibrated, the ensemble is stacked on a later chronological stage, and a final isotonic calibration is fitted on a still-later validation stage before the locked test is opened.

The outer split uses chronological purge and a 252-business-day embargo. Training observations span January 1990 through September 2004; the three internal validation roles span October 2005 through February 2014; and the locked test spans March 2015 through June 2021. Earlier-stage rows are retained only when their target end date is strictly before the next fitting stage begins.

All 14 configured gate checks pass, including minimum sample and class counts, temporal breadth, ROC AUC, proper-score skill, ECE, calibration slope, walk-forward stability, and the dependence-aware bootstrap checks. The moving date-block/cross-sectional cluster bootstrap uses 100 draws with 12-date blocks. Its 90%

Table 5: Locked-test results for the real historical 12-month research model. The table separates discrimination, calibration, and proper-score improvement.

Metric	Value
ROC AUC	0.576221
Average precision	0.693812
Brier score	0.218250
Brier skill	0.036425
Log loss	0.628046
Log-loss skill	0.026983
ECE (10 bins)	0.044712
Calibration slope	1.272327
Calibration intercept	-0.028144
Event rate	0.656015
Locked-test observations	532
Distinct observation dates	76
Test span	2,283 days

interval for ROC AUC is approximately $[0.498, 0.602]$; for Brier skill, approximately $[-0.0035, 0.0562]$; for log-loss skill, approximately $[-0.0030, 0.0424]$; and for ECE, approximately $[0.0167, 0.0910]$.

These intervals are an important part of the result. The point estimates are positive, but the resampled lower bounds for AUC and proper-score skill are close to the configured acceptance floor. The artifact is therefore evidence that the pipeline can produce a statistically eligible probability model on real historical data; it is not evidence of a large or guaranteed economic edge.

The artifact is assigned `validated_research`. Promotion to `validated_market` is intentionally blocked because the training sample does not establish historical-universe survivorship control, inclusion of delisted securities, or a documented corporate-action-adjustment policy. Public redistribution of the trained artifact is also held behind an upstream data/model-rights review. No raw training rows are required for inference, but absence of raw rows is not treated as automatic permission to redistribute a derived model.

At runtime, ordinary observed daily price histories are aggregated to month-end before the 20-feature contract is reconstructed. This preserves the monthly training semantics instead of forward-filling monthly data into fabricated daily observations. The model is installed as an eligible candidate but is not automatically activated. Forecast selection and Live activation remain explicit user decisions.

Other requested horizons remain governed by the same principle. Six-month candidates did not satisfy every configured gate; two-year candidates remained below required skill conditions; and the three-year target did not have adequate chronological support for promotion under the multi-stage leakage-safe validation design used here. These horizons are therefore not represented as passing models.

19 A complete worked example

It is helpful to follow one hypothetical security through the three planes.

Assume first that the evidence engine observes strong returns on capital, acceptable leverage, a valuation near the sector median, and incomplete macro data. The normalized evidence values are aggregated through Equation (15). The resulting composite may be, for example, 7.6 with a broad model interval because the macro and one balance-sheet field are missing. This output says “the evidence profile is relatively strong, but some evidence is absent." It does not yet say anything like "76% probability of outperformance."

Now suppose the frozen anchor model, using the exact feature set and target for which it was validated, produces

$$p_t^A = 0.58. \tag{59}$$

This number has an empirical meaning because the anchor is evaluated under proper scores and calibration diagnostics on held-out data. It still does not mean that the future is known. It means only that 0.58 is the model's probability estimate for the event defined in Equation (3), conditional on the information contract.

Later the same day a fresh filing arrives and the stock moves sharply relative to the benchmark. Suppose the adaptive feature vector and posterior mean produce

$$z_t^\top m_t = 0.33. \tag{60}$$

Because $0.33 < d_{\max} = 0.75$, the bound does not alter the shift. The candidate becomes

$$p_t^C = \text{sigmoid}\{\text{logit}(0.58) + 0.33\} \approx 0.658. \tag{61}$$

If the adaptive gate is active and the relevant sources are verified as fresh, Equation (58) allows 0.658 to become the live probability. If the gate is warming, degraded, or in drift alert, the live output remains 0.58.

Nothing is learned from the future at this point. The pending record stores the observation date, exact target definition, benchmark, anchor probability, adaptive candidate, feature vector and state lineage. At the 252nd common trading session, the target is resolved. Only then does the model calculate the anchor and adaptive losses, update the gate and drift state, and apply Equations (48)–(49) for subsequent forecasts.

This worked example illustrates the central principle: *new information may alter a belief before it is allowed to alter the model that learns from realized outcomes*.

20 What counts as market validation?

The real historical 12-month artifact establishes a research-level result, but the stronger market claim still requires a dataset whose information chronology and security-universe construction are controlled at a higher level. The next scientific step is therefore not simply to add another learner to the ensemble. It is to construct a market-grade historical dataset for which the locked test means what it is supposed to mean.

At minimum, a market-validation dataset should satisfy the following conditions. Historical features must use actual availability times, including filing acceptance times where relevant. Macroeconomic variables must be reconstructed from historical vintages rather than revised present-day values. The security universe must be historical rather than reconstructed from today's survivors. Delisted securities and their terminal returns must be represented. Prices must be adjusted under a documented corporate-action policy. The benchmark and stock price calendars must support the exact common-session target endpoint. Provider transformations and missing-data rules must be fixed before the confirmatory test is opened.

The research process itself also needs governance. If dozens of feature sets, horizons, thresholds, calibrators and model classes are repeatedly compared on the same "locked" period, the test period has become part of the search procedure. The central lesson of backtest-overfitting research is that selection history matters (Bailey et al., 2017). A serious confirmatory study should therefore pre-register the target, feature contract, validation gates and analysis plan before opening a new locked period.

The existing 12-month artifact demonstrates the first of these two levels: its statistical gate passes on real historical data and therefore supports `validated_research`. The stronger `validated_market` designation additionally requires documented controls for point-in-time features, survivorship, delistings, and corporate actions. Neither tier guarantees future profitability. They describe how much evidence exists that the probability model behaved as claimed under a defined historical protocol.

21 Limitations

Several limitations are deliberate.

First, the evidence-plane Beta distribution is a model of rubric uncertainty. The metric weights and transformations are structured judgments, not frequencies learned from realized returns. Its posterior probabilities must therefore remain inside the evidence-score interpretation.

Second, the anchor uncertainty envelope is not a fully Bayesian posterior predictive distribution over all model and market uncertainty. The logistic component has a coefficient posterior, whereas the tree components contribute model dispersion through their point probabilities. The final interval is useful for caution and abstention, but it has no claimed frequentist coverage guarantee for realized returns.

Third, the adaptive residual is local and bounded. This is a safety feature, but it also limits how quickly the system can adapt after a genuine regime break. A larger process noise or smaller forgetting factor can make the residual more responsive, but those choices create a different state lineage and require their own validation.

Fourth, the drift rule in Equation (57) is a monitoring heuristic, not a calibrated statistical test. It is intended to suppress a deteriorating online layer, not to diagnose the economic cause of a regime change.

Fifth, the current live event vector records filing recency and filing family, not the semantic content of the filing. Adding textual event models, news, options, order-book or analyst-revision signals may be useful, but each new source changes the information contract, latency, licensing, validation burden and potential leakage paths.

Finally, the synthetic fixture results are not evidence of market alpha. The real historical 12-month result is stronger evidence than a fixture because it passed the configured research gate on real observations, but it remains limited by the underlying surviving-sample construction and unresolved delisting/corporate-action controls. Its bootstrap intervals are also materially wider than the point estimates alone might suggest. The appropriate claim is therefore calibrated research eligibility, not durable alpha.

22 Discussion

Financial forecasting systems are often described as though the main problem were choosing the most powerful learner. In practice, the more difficult problem is preserving the meaning of a probability while data, models and market conditions change.

The architecture in this paper treats the frozen model as an anchor rather than as a permanent truth. That choice has two consequences. It allows new evidence to matter quickly, because the residual can move the candidate probability without waiting for a complete retraining cycle. At the same time, it gives the system somewhere safe to return when the adaptive layer has not accumulated enough independent dates or begins to underperform. The anchor is not immune to regime change, but replacing a validated model with an unvalidated online state is not an improvement merely because the replacement is newer.

The use of log-odds is also more than algebraic convenience. A bounded additive correction in log-odds space corresponds to a bounded multiplicative change in odds. This makes the influence of the live layer easier to reason about than an unrestricted probability offset, whose effect depends strongly on whether the anchor is near 0.5 or near a boundary.

The most important methodological separation remains the first one: evidence uncertainty is not forecast uncertainty. There is value in a transparent Bayesian score even when no market-probability model has been validated. There is also value in a calibrated empirical probability even when the explanatory story is incomplete. Combining the two by relabeling one as the other would make both less informative.

23 Conclusion

A financial probability is meaningful only relative to a precise event, a time-specific information set and a validation record. Once those conditions are taken seriously, the architecture of a live forecasting system changes. Static historical validation remains necessary, but it is no longer sufficient. New information must be incorporated without hindsight, online learning must wait for outcomes to mature, repeated observations must not manufacture temporal evidence, and an adaptive model must be able to lose control without taking the validated anchor down with it.

The model developed here implements those requirements as a sequence of explicit statistical contracts. An interpretable evidence posterior is kept separate from an empirical forward probability. The forward model is trained and calibrated under purge-and-embargo chronology and evaluated on a locked test with dependence-aware uncertainty. A bounded Bayesian residual updates the candidate log-odds from fresh events. Its posterior learns only after exact-horizon labels mature, and its live use depends on date-balanced prequential loss, calibration, temporal breadth, freshness and drift checks. Artifact hashes and settings fingerprints make these claims reproducible at the software level.

The synthetic fixtures show that the machinery can be exercised end to end without being confused with market evidence. The real historical 12-month artifact adds a second result: a calibrated ensemble can pass the configured locked-test research gate while still being prevented from making a stronger market claim because the dataset lacks the required historical-universe controls. That distinction is not a weakness of the framework; it is the point. The next credible escalation is a confirmatory study on a genuinely point-in-time, survivorship-controlled, delisting-aware, corporate-action-adjusted market dataset under a protocol fixed before the new locked period is examined.

A Selected derivations

A.1 Moments of the evidence posterior

For $U \sim \mathrm{Beta}(\alpha, \beta)$,

$$\mathbb{E}[U] = \frac{\alpha}{\alpha+\beta}, \qquad \mathrm{Var}(U) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}. \tag{62}$$

Scaling by ten multiplies the mean by ten and the variance by one hundred. These identities explain why the evidence score shrinks toward five under the symmetric prior and why additional weighted evidence narrows the posterior.

A.2 Gradient and Hessian for Bayesian logistic regression

Let $p = \mathrm{sigmoid}(X\beta)$ and place the prior in Equation (22). The gradient of the negative log posterior is

$$\nabla \mathcal{L}(\beta) = X^\top (p - y) + \sigma_\beta^{-2}\beta, \tag{63}$$

and the Hessian is

$$\nabla^2 \mathcal{L}(\beta) = X^\top W X + \sigma_\beta^{-2} I. \tag{64}$$

At the MAP estimate, the inverse Hessian gives the Laplace covariance in Equation (25).

A.3 Why the online covariance update is rank one

Before the matured Bernoulli observation is incorporated, let the Gaussian covariance be P^-. The local negative Hessian contribution of one logistic observation is $wzz^\top$, so the approximate posterior precision is

$$(P^+)^{-1} = (P^-)^{-1} + wzz^\top. \tag{65}$$

Using the Sherman-Morrison formula,

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u}, \tag{66}$$

with $A = (P^-)^{-1}$ and $u = v = \sqrt{w}z$ gives

$$P^+ = P^- - \frac{wP^- zz^\top P^-}{1 + wz^\top P^- z}, \tag{67}$$

which is Equation (48) after defining $v = P^- z$ and $d = 1 + wz^\top v$.

A Newton-style mean correction uses the Bernoulli score $(y - p)z$ and the same local curvature, yielding

$$m^+ = m^- + \frac{P^- z}{1 + wz^\top P^- z}(y - p). \tag{68}$$

The update is approximate because the logistic likelihood is not conjugate to the Gaussian prior; it is a sequential local Laplace approximation.

A.4 Odds interpretation of the adaptive shift

Let $O(p) = p/(1 - p)$ denote odds. From Equation (39) before probability clipping,

$$\log O(p_t^C) = \log O(p_t^A) + \delta_t, \tag{69}$$

$$O(p_t^C) = O(p_t^A)e^{\delta_t}. \tag{70}$$

Thus the bound $|\delta_t| \leq d_{\max}$ directly limits the multiplicative change in odds. This property is one reason the residual is expressed in log-odds rather than as $p_t^A + \Delta p_t$.

B Reference configuration

Table 6 records the installed 12-month research-anchor settings that are directly relevant to the reported result. Table 7 lists the balanced adaptive profile. These values document the implementation used for the examples; they are not universal statistical recommendations.

C Software invariants

The reference implementation encodes several statistical assumptions as testable software invariants. The most important are:

1. A feature cannot be joined into a historical observation before its documented availability time.
2. The forward label resolves at the exact Hth common stock/benchmark trading session, not at the scheduler's processing date.
3. The locked test is excluded from model fitting, component calibration, stacking and final calibration.
4. Outer train/validation/test boundaries purge unresolved target windows and apply the configured embargo; internal validation-role boundaries purge unresolved targets without reapplying the full outer embargo.
5. The adaptive state is fingerprinted by the base model, engine identifier and complete adaptive settings contract.
6. Pending labels carry the same lineage as the state that generated them.

Table 6: Selected settings for the 12-month research anchor.

Setting	Value
Horizon	12 months / 252-session target
Feature frequency	Monthly
Benchmark	S&P 500 index
Excess-return hurdle	0
Sampling step	21 trading days
Train / validation / test	0.50 / 0.30 / 0.20
Outer embargo	252 business days
Bayesian prior standard deviation	1.5
Component/final calibration	Isotonic
Posterior draws	1,200
Credible level	0.90
Bootstrap draws	100
Bootstrap date block	12 dates
Walk-forward folds	3
Minimum test rows	150
Minimum test dates	9
Minimum test span	180 days
Maximum ECE	0.15
Calibration slope range	[0.25,2.5]
Minimum ROC AUC	0.50
Minimum Brier skill	-0.02
Minimum log-loss skill	-0.02

Table 7: Selected adaptive settings used by the reference implementation.

Setting	Value
Adaptive prior standard deviation	0.75
Maximum log-odds shift	0.75
Forgetting factor	0.997
Process noise	0.0005
Probability clip	0.01
Event half-life	36 hours
Market refresh cadence	60 s
SEC refresh cadence	600 s
Macro refresh cadence	900 s
Market verification staleness limit	900 s
SEC verification staleness limit	86,400 s
Macro verification staleness limit	86,400 s
Minimum matured observations	120
Minimum unique dates	60
Minimum observation span	180 days
Evaluation window	250 observation dates
Maximum ECE	0.08
Maximum Brier regret	0
Maximum log-loss regret	0
Drift EWMA α	0.08
Drift control multiplier	2.75
Drift minimum samples	60

7. Repeated browser refreshes do not create independent matured learning evidence.
8. Online Brier score, log loss and ECE use date-balanced weighting for gate decisions.
9. Provider-verification age is distinct from event age; stale verification suppresses source-dependent adaptive features.
10. The adaptive residual cannot make an ineligible frozen anchor live.
11. A failed adaptive gate returns live control to the frozen anchor.
12. Synthetic fixtures remain permanently ineligible for live market forecasting.

These controls do not guarantee a profitable model. They do make several common forms of accidental hindsight, metric inflation and uncontrolled online adaptation harder to introduce without breaking a test or changing a model identity.

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